Problem M. Rejection Sampling

Time limit: 1 second
Memory limit: 1024 megabytes

Bobo wants to use a rejection sampling algorithm to construct a random set $T \subset \{1, 2, \ldots, n\}$ of size k. For parameters $p_1, p_2, ..., p_n$ $(0 \le p_i \le 1)$ and integer k, the rejection sampler is defined as follows:

1. Initialize $T \leftarrow \emptyset$;
2. For each i $(1 \le i \le n)$, add i into T with probability p_i;
3. Output T if the size of T is exactly k; otherwise, repeat the process.

Now you are given integers $a_1, a_2, ..., a_n$ and k. Bobo needs to set the parameters $p_1, p_2, \ldots, p_n$ satisfying

- $\sum_{i=1}^{n} p_i = k$;
- for all $S \subseteq \{1, 2, \cdots, n\}$ such that $|S| = k$, the probability that the rejection sampler outputs S is proportional to $\prod_{i \in S} a_i$.

Your task is to find out the parameters $p_1, p_2, \ldots, p_n$ for Bobo. It is guaranteed that such parameters exist and **are unique**. Your answer will be considered correct if the absolute error of each p_i doesn't exceed 10^{-6} compared to the unique answer.

Input

The first line of the input contains two integers n and k $(2 \le n \le 10^5, 1 \le k \le n - 1)$.

The second line of the input contains n integers $a_1, a_2, \ldots, a_n$ $(1 \le a_i \le 10^9)$.

Output

Output n lines. The i-th line contains a single real number p_i.

Your answer is considered correct if the absolute error of each parameter does not exceed 10^{-6}. Namely, if your answer is a, and the jury's answer is b, then your answer is accepted if $|b - a| \le 10^{-6}$ for all parameters.

Examples

standard input	standard output
3 2 5 5 5	0.666666666667 0.666666666667 0.666666666667
2 1 1 4	0.333333333333 0.666666666667
4 2 1 2 3 4	0.310035697652 0.473324044845 0.574114878920 0.642525378583